Tutorial 9: Linear Classifiers

Four problems in classification and learning

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Semester A, 2026/27

Task 1: Least-squares classification

A quality-control sensor produces five labeled readings:

Reading x_i	-2	-1	0	1	2
Label y_i	-1	-1	-1	$+1$	$+1$

Fit $f(x) = wx + b$ by minimizing $\sum_i (y_i - wx_i - b)^2$. Use the augmented design matrix with rows $(x_i, 1)$ and $\boldsymbol{\beta} = (w, b)^\mathsf{T}$.

(a) Form $\boldsymbol{X}$ and compute $\boldsymbol{X}^\mathsf{T}\boldsymbol{X}$ and $\boldsymbol{X}^\mathsf{T}\boldsymbol{y}$.

(b) Solve for w, b and give the decision boundary.

(c) Compute the scores and predicted classes for $x = 1/2$ and $x = 1/4$.

(d) A new positive example has $x = 4$, $y = +1$. Compute its squared loss and explain why it is nonzero even if the class is correct.

Solution 1(a): Normal-equation terms

(a) Construct the augmented data and compute the sums. With the feature column followed by a column of ones:

$$\boldsymbol{X} = \begin{bmatrix} -2 & 1 \\ -1 & 1 \\ 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix}, \qquad \boldsymbol{y} = \begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \\ 1 \end{bmatrix}.$$

The four sums are:

$$\sum_i x_i = 0, \qquad \sum_i x_i^2 = 4 + 1 + 0 + 1 + 4 = 10,$$
$$\sum_i y_i = -1, \quad \sum_i x_i y_i = 2 + 1 + 0 + 1 + 2 = 6.$$

Therefore:

$$\boxed{\boldsymbol{X}^\top \boldsymbol{X} = \begin{bmatrix} 10 & 0 \\ 0 & 5 \end{bmatrix}}, \qquad \boxed{\boldsymbol{X}^\top \boldsymbol{y} = \begin{bmatrix} 6 \\ -1 \end{bmatrix}}.$$

Solution 1(b): Fitted classifier

(b) Solve the normal equations and find the boundary. Least squares requires $\boldsymbol{X}^\top \boldsymbol{X}\boldsymbol{\beta} = \boldsymbol{X}^\top \boldsymbol{y}$:

$$\begin{bmatrix} 10 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} w \\ b \end{bmatrix} = \begin{bmatrix} 6 \\ -1 \end{bmatrix}.$$

Thus $10w = 6$ and $5b = -1$, giving:

$$\boxed{w = 0.6, \qquad b = -0.2}, \qquad \boxed{f(x) = 0.6x - 0.2}.$$

The decision boundary satisfies $f(x) = 0$:

$$0.6x - 0.2 = 0 \quad \Longrightarrow \quad \boxed{x = \frac{1}{3}}.$$

Since the readings have mean zero, $b = \bar{y} = -1/5$. The class-label imbalance shifts this boundary away from zero.

Solution 1(c): Predictions

(c) Classify by the sign of the fitted score. For $x = 1/2$:

$$f(1/2) = 0.6(1/2) - 0.2 = 0.10 > 0 \quad \Longrightarrow \quad \boxed{\hat{y} = +1}.$$

For $x = 1/4$:

$$f(1/4) = 0.6(1/4) - 0.2 = -0.05 < 0 \quad \Longrightarrow \quad \boxed{\hat{y} = -1}.$$

Both observations are near the boundary $x = 1/3$. Their predictions differ because they are on opposite sides. The real-valued score is not a class probability.

Solution 1(d): Squared-loss limitation

(d) Evaluate the new positive example $(x, y) = (4, +1)$. The classifier produces:

$$f(4) = 0.6(4) - 0.2 = 2.2 > 0.$$

It predicts the positive class correctly. However, its squared loss is:

$$\boxed{\ell = (y - f(4))^2 = (1 - 2.2)^2 = 1.44}.$$

Least squares penalizes deviation from the numerical target $+1$, even for a correct classification. Larger positive scores do not always mean smaller squared loss.
This motivates classification-specific objectives that focus on which side of the boundary an observation lies.

Task 2: Perceptron learning

An online classifier processes four observations in the order given. Each input has two features, $\boldsymbol{x}_i = (u_i, v_i)^\mathsf{T}$:

i	(u_i, v_i)	y_i
1	$(1, 0)$	$+1$
2	$(2, 2)$	-1
3	$(0, -1)$	$+1$
4	$(-1, 2)$	-1

Start with $\boldsymbol{w} = (0, 0)^\mathsf{T}$, $b = 0$, and $\eta = 1$. Use $f(\boldsymbol{x}) = \boldsymbol{w}^\mathsf{T}\boldsymbol{x} + b$. When $y_i f(\boldsymbol{x}_i) \le 0$, update $\boldsymbol{w} \leftarrow \boldsymbol{w} + \eta y_i \boldsymbol{x}_i$ and $b \leftarrow b + \eta y_i$.

(a) Compute the first signed score and update the parameters.

(b) Process the second example using the current parameters.

(c) Process examples 3 and 4, then state the final boundary.

(d) Reclassify all four examples. Explain whether one pass is sufficient and give a possible separator for comparison.

Solution 2(a): First update

(a) Process $\boldsymbol{x}_1 = (1, 0)^\mathsf{T}$ with $y_1 = +1$. At initialization $\boldsymbol{w} = (0, 0)^\mathsf{T}$ and $b = 0$:

$$f(\boldsymbol{x}_1) = 0, \qquad y_1 f(\boldsymbol{x}_1) = 0.$$

By the stated convention, a zero signed score triggers an update:

$$\boldsymbol{w}^{(1)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + (+1) \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \boxed{\begin{bmatrix} 1 \\ 0 \end{bmatrix}},$$

$$b^{(1)} = 0 + 1 = \boxed{1}.$$

The update increases the positive example's score. We must use these new parameters when processing the next observation.

Solution 2(b): Second update

(b) Process $\boldsymbol{x}_2 = (2,2)^\mathsf{T}$ with $y_2 = -1$. With $\boldsymbol{w}^{(1)} = (1,0)^\mathsf{T}$ and $b^{(1)} = 1$:

$$f(\boldsymbol{x}_2) = 1(2) + 0(2) + 1 = 3, \quad y_2 f(\boldsymbol{x}_2) = (-1)(3) = -3 < 0.$$

The negative example lies on the positive side, so update:

$$\boldsymbol{w}^{(2)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} + (-1) \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \boxed{\begin{bmatrix} -1 \\ -2 \end{bmatrix}},$$

$$b^{(2)} = 1 - 1 = \boxed{0}.$$

An update corrects the current example's score direction, but it may change the scores of previously seen examples.

Solution 2(c): Remaining observations

(c) Use $\boldsymbol{w}^{(2)} = (-1, -2)^{\mathsf{T}}$, $b^{(2)} = 0$. Example 3 has $\boldsymbol{x}_3 = (0, -1)^{\mathsf{T}}$, $y_3 = +1$:

$$f(\boldsymbol{x}_3) = -1(0) - 2(-1) = 2, \quad y_3 f(\boldsymbol{x}_3) = 2 > 0.$$

Example 4 has $\boldsymbol{x}_4 = (-1, 2)^{\mathsf{T}}$, $y_4 = -1$:

$$f(\boldsymbol{x}_4) = -1(-1) - 2(2) = -3, \quad y_4 f(\boldsymbol{x}_4) = 3 > 0.$$

Both signed scores are positive, so neither observation triggers an update. The final score and boundary, for $\boldsymbol{x} = (u, v)^{\mathsf{T}}$, are:

$$\boxed{f(u, v) = -u - 2v}, \qquad \boxed{-u - 2v = 0}.$$

Solution 2(d): Final signed scores

(d) Verify all four examples after the first pass. Using $f(u, v) = -u - 2v$:

i	y_i	$f(\boldsymbol{x}_i)$	$y_i f(\boldsymbol{x}_i)$	Correct?
1	$+1$	-1	-1	No
2	-1	-6	$+6$	Yes
3	$+1$	$+2$	$+2$	Yes
4	-1	-3	$+3$	Yes

The final classifier correctly labels 3 of 4 training examples. The first point, initially corrected, became misclassified after processing the second point.
Thus a single pass need not be sufficient, even when a separating line exists. The next page gives one such line.

Solution 2(d): One-pass limitation

(d) A separating line exists despite the one-pass error. For example, consider the score $f_{\text{sep}}(u, v) = 1 - v$:

Example	v	$f_{\text{sep}} = 1 - v$
1	0	$+1$
2	2	-1
3	-1	$+2$
4	2	-1

Its signs agree with all four training labels, so $v = 1$ is a valid separating boundary.
The perceptron *can* converge on linearly separable data, but a particular sample order does not guarantee that one pass already reaches a correct separator. Continue iterating until an entire pass requires no updates.

Task 3: Logistic training step

A binary defect classifier uses three two-feature observations:

i	$\boldsymbol{x}_i^\mathsf{T}$	y_i
1	$(2, 2)$	1
2	$(-1, 2)$	0
3	$(1, -2)$	1

Let $p_i = \sigma(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_i + b)$, where $\sigma(a) = 1/(1 + e^{-a})$. Start from $\boldsymbol{w} = (0, 0)^\mathsf{T}$, $b = 0$ and use $\eta = 0.6$. Minimize average binary cross-entropy with an ℓ_2 penalty $J_\lambda = \mathrm{BCE}_{\mathrm{avg}} + \lambda\|\boldsymbol{w}\|_2^2/2$, where $\lambda = 0.3$ and the intercept is not penalized.

(a) Compute the initial probabilities and J_λ.

(b) Compute the weight and intercept gradients at initialization.

(c) Perform one full-batch gradient-descent update.

(d) Recompute the probabilities and J_λ. Compare with part (a).

Solution 3(a): Initial loss

(a) Evaluate the initial probability model. With zero weights and intercept, every linear score is zero:

$$a_1 = a_2 = a_3 = 0, \qquad \boxed{p_1 = p_2 = p_3 = \sigma(0) = 0.5}.$$

For label $y_i \in \{0, 1\}$, one binary cross-entropy term is:

$$\ell_i = -y_i \log p_i - (1 - y_i) \log(1 - p_i).$$

Each example therefore has loss $-\log 0.5 = \log 2$. The penalty is initially zero because $\boldsymbol{w} = \mathbf{0}$.

$$\boxed{J_\lambda^{(0)} = \log 2 \approx 0.6931}.$$

This is an uninformative starting model: the positive-class probability is 0.5 for every example.

Solution 3(b): Gradient

(b) Compute the full-batch gradient at initialization. For $n = 3$, average BCE with ℓ_2 regularization gives:

$$\nabla_{\boldsymbol{w}} J_\lambda = \frac{1}{3}\sum_{i=1}^{3}(p_i - y_i)\boldsymbol{x}_i + \lambda \boldsymbol{w},$$

$$\frac{\partial J_\lambda}{\partial b} = \frac{1}{3}\sum_{i=1}^{3}(p_i - y_i).$$

At initialization, $(p_1 - y_1, p_2 - y_2, p_3 - y_3) = (-0.5, 0.5, -0.5)$ and $\lambda \boldsymbol{w} = \mathbf{0}$.

$$\nabla_{\boldsymbol{w}} J_\lambda = \frac{1}{3}\left[-0.5\begin{bmatrix}2\\2\end{bmatrix} + 0.5\begin{bmatrix}-1\\2\end{bmatrix} - 0.5\begin{bmatrix}1\\-2\end{bmatrix}\right] = \boxed{\begin{bmatrix}-2/3\\1/3\end{bmatrix}},$$

$$\frac{\partial J_\lambda}{\partial b} = \frac{-0.5 + 0.5 - 0.5}{3} = \boxed{-1/6}.$$

Solution 3(c): Parameter update

(c) Take one gradient step with $\eta = 0.6$. Gradient descent subtracts η times each gradient:

$$\boldsymbol{w}^{(1)} = \mathbf{0} - 0.6 \begin{bmatrix} -2/3 \\ 1/3 \end{bmatrix} = \boxed{\begin{bmatrix} 0.4 \\ -0.2 \end{bmatrix}},$$

$$b^{(1)} = 0 - 0.6(-1/6) = \boxed{0.1}.$$

The positive weight on feature 1 and negative weight on feature 2 are determined by their combined contributions across all three labels.
We next evaluate the *new* probabilities rather than assuming that one gradient step must always improve the loss.

Solution 3(d): Updated probabilities

(d) Recompute the probabilities using the new parameters. With $\boldsymbol{w}^{(1)} = (0.4, -0.2)^\mathsf{T}$ and $b^{(1)} = 0.1$:

$$a_1 = 0.4(2) - 0.2(2) + 0.1 = 0.5,$$
$$a_2 = 0.4(-1) - 0.2(2) + 0.1 = -0.7,$$
$$a_3 = 0.4(1) - 0.2(-2) + 0.1 = 0.9.$$

The corresponding positive-class probabilities are:

$$\boxed{(p_1, p_2, p_3) = (\sigma(0.5), \sigma(-0.7), \sigma(0.9)) \approx (0.6225, 0.3318, 0.7109)}.$$

The model now assigns a higher positive-class probability to both positive examples and a lower one to the negative example.

Solution 3(d): Updated objective

(d) Evaluate the new loss, including regularization. For labels $(1, 0, 1)$, the average binary cross-entropy is:

$$\begin{aligned} \mathrm{BCE}_{\mathrm{avg}} &= -\frac{1}{3}[\log p_1 + \log(1 - p_2) + \log p_3] \\ &\approx \frac{0.4741 + 0.4032 + 0.3412}{3} = \boxed{0.4061}. \end{aligned}$$

The ℓ_2 penalty uses $\lambda = 0.3$:

$$\frac{\lambda}{2}\|\boldsymbol{w}^{(1)}\|_2^2 = \frac{0.3}{2}(0.4^2 + (-0.2)^2) = \boxed{0.0300}.$$

Thus:

$$\boxed{J_\lambda^{(1)} \approx 0.4361 < 0.6931 = J_\lambda^{(0)}}.$$

Because $\lambda \boldsymbol{w} = \mathbf{0}$ at initialization, regularization first affects later gradient steps, once the weights are nonzero.

Task 4: Multiclass softmax

A text classifier uses three linear scores $a_k = \boldsymbol{w}_k^\top \boldsymbol{x} + b_k$ for categories A, B, C. For one embedding $\boldsymbol{x}_* = (2, 1)^\top$, its parameters are:

Class k	$\boldsymbol{w}_k^\top$	b_k
A	$(1, -1)$	0
B	$(0, 1)$	-1
C	$(1, 1)$	0

The observed label is B. Use $p_k = e^{a_k} / \sum_j e^{a_j}$, loss $\ell = -\log p_B$, and a bias-only learning rate $\eta = 1$ (all $\boldsymbol{w}_k$ stay fixed).

(a) Compute the three scores and the current predicted class.

(b) Compute the softmax probabilities and ℓ, subtracting the maximum score before exponentiation.

(c) Derive $\partial \ell / \partial a_k$ for all three scores.

(d) Update the three biases once. Recompute p_B and describe the effect on the observed class.

Solution 4(a): Class scores

(a) Compute the scores before softmax. Use $a_k = \boldsymbol{w}_k^{\mathsf{T}} \boldsymbol{x}_* + b_k$ with $\boldsymbol{x}_* = (2, 1)^{\mathsf{T}}$:

$$
\begin{aligned}
a_A &= 1(2) - 1(1) + 0 = 1, \\
a_B &= 0(2) + 1(1) - 1 = 0, \\
a_C &= 1(2) + 1(1) + 0 = 3.
\end{aligned}
$$

Therefore:

$$\boxed{(a_A, a_B, a_C) = (1, 0, 3)}.$$

Since a_C is the largest, the current prediction is $\boxed{C}$, although the observed label is B.
The scores are not probabilities; softmax converts them into three nonnegative values that sum to one.

Solution 4(b): Softmax probabilities

(b) Normalize the class scores. Subtract the maximum score 3, giving $(-2, -3, 0)$. This shift leaves all softmax probabilities unchanged:

$$(p_A, p_B, p_C) = \frac{(e^{-2}, e^{-3}, 1)}{e^{-2} + e^{-3} + 1}$$
$$\approx \boxed{(0.1142,\ 0.0420,\ 0.8438)}.$$

The correct class B receives only about 4.2% probability. Its negative log-likelihood is:

$$\boxed{\ell = -\log p_B \approx -\log(0.0420) \approx 3.1698}.$$

A confidently wrong prediction receives a large cross-entropy loss.

Solution 4(c): Softmax gradient

(c) Differentiate the loss with respect to each score. Let $Z = \sum_{j\in\{A,B,C\}} e^{a_j}$. For observed class B:

$$\ell = -\log \frac{e^{a_B}}{Z} = -a_B + \log Z.$$

Differentiate with respect to a_k:

$$\frac{\partial \ell}{\partial a_k} = -\mathbb{I}[k = B] + \frac{e^{a_k}}{Z} = \boxed{p_k - \mathbb{I}[k = B]}.$$

Here $\mathbb{I}[k = B]$ equals 1 for class B and 0 otherwise. Thus, in class order (A, B, C):

$$\boxed{\nabla_{\boldsymbol{a}} \ell = (p_A, p_B - 1, p_C) \approx (0.1142, -0.9580, 0.8438)}.$$

Solution 4(d): Bias update

(d) Update only the three class biases. Because $a_k = \boldsymbol{w}_k^{\mathsf{T}} \boldsymbol{x}_* + b_k$, $\partial a_k / \partial b_k = 1$. The bias gradients equal the score gradients from part (c).
With $\eta = 1$, use $b_k^{\text{new}} = b_k - \partial \ell / \partial a_k$:

$$
\begin{aligned}
b_A^{\text{new}} &= 0 - 0.1142 \approx -0.1142, \\
b_B^{\text{new}} &= -1 - (-0.9580) \approx -0.0420, \\
b_C^{\text{new}} &= 0 - 0.8438 \approx -0.8438.
\end{aligned}
$$

The update raises the score of the observed class B relative to both competing classes.

Solution 4(d): Updated prediction

(d) Re-evaluate the probability of the observed class. Since the weight vectors are fixed, only the biases change:

$$(a_A^{\text{new}}, a_B^{\text{new}}, a_C^{\text{new}}) \approx (0.8858, 0.9580, 2.1562).$$

Applying softmax again gives:

$$\boxed{(p_A^{\text{new}}, p_B^{\text{new}}, p_C^{\text{new}}) \approx (0.1774, 0.1907, 0.6319)}.$$

The true-class probability increases:

$$\boxed{p_B : \ 0.0420 \ \longrightarrow \ 0.1907}.$$

The predicted class is still C, but the loss decreases from 3.1698 to 1.6572. A higher true-class probability need not immediately change the predicted label.

Tutorial 9 complete

Questions and discussion